

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Specialist techniques & statistical analysis - Set 1

Mark scheme

AQA spec 3.7.2.3; Research methods; textbook pp. 402-407, 420-423

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Image databases: identifying individuals (tiger stripes, whale shark spots, dolphin fins) to find territory size, movements, lifespan, social groups - few species have unique recognisable features • eDNA: detecting the presence of a species from DNA shed into water (great crested newt, invasive carp) - shows presence not abundance; contamination / DNA transport by water • Geolocator tags: tracking small birds on migration where transmitters are too heavy - only record light and time so position is approximate; animal must be recaptured to download data	6
01.2	• Microwaves emitted by the satellite reflect off the surface giving data on wave height, wind velocity, sea surface altitude and oil slicks (which smooth the surface) • Microwaves pass through cloud so monitoring is not interrupted by weather	2
01.3	<i>Any 2 from:</i> • Droppings: diet, gender, territory size • Owl pellets: diet • Feeding marks (chewed hazel nuts - dormouse), tracks / footprints, nests / burrows, scratching posts (territory)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-407; AQA spec 3.7.2.3.	
02.1	• The scatter / variability of results around each mean; about 68% of values lie within one SD of the mean • The SDs are large relative to the difference in means (1.9 g) so the ranges overlap greatly	2
02.2	• t-test • Comparing two means of measured (continuous) data	2

Question	Marking guidance	Marks
02.3	• There is a 30% probability that the difference occurred by random chance • Above the 0.05 (5%) level so the difference is not significant; the null hypothesis (no difference) is accepted	2
02.4	• Two correlated variables may both be caused by a third variable that was not investigated • Significance only shows the result is unlikely to be chance; a mechanism / controlled experiment is needed for causation	2
02.5	• Increase the number of replicates (reduces the effect of variability) • Standardise other variables (soil, water, light) more closely to reduce scatter	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 420-423; AQA spec 3.7.2.3.	
03.1	• Spearman's rank correlation coefficient • Chi-squared test (frequencies / counts in categories) • t-test (two means of measured data) • Spearman's rank correlation coefficient	4
03.2	• Chi-squared compares frequencies / counts in categories; it cannot be used for continuously variable measured data	1
03.3	• The value (ignoring sign) exceeds the critical value so the correlation is significant at the 5% level • The negative sign shows a negative correlation: one variable decreases as the other increases • There is less than a 5% probability the correlation arose by chance; reject the null hypothesis (but correlation is not proof of causation)	3
03.4	• Show the standard deviation / range of the original values around each mean • If the bars of two means overlap greatly the difference is probably not significant; little overlap suggests a significant difference	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 422-423; AQA spec 3.7.2.3.	
04.1	• Each regional population has its own unique gene pool / DNA profile • DNA from seized timber or ivory is compared with the database of regional populations • This identifies where the material came from, showing whether it was legally harvested and where poaching occurs	3
04.2	• Radio / GPS transmitters use radio waves - work in air and fresh water, giving positions (GPS stores or transmits location) • Radio waves do not travel through seawater so acoustic transmitters (sound) are used for marine animals, giving real-time positions	2
04.3	• Drones / ROVs, AUVs (eg Autosub), balloons, aircraft, satellites, animals (tags) • Drones: cheap, low disturbance, high-resolution images of areas hard to reach (cliffs, canopy, ice), repeated surveys; infrared cameras detect animals at night	3
04.4	• Cameras record automatically (visible light by day, infrared at night) without researchers being present • So animals behave naturally and nests are not deserted; monitoring continues 24 hours	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 404-406; AQA spec 3.7.2.3.	
05.1	• $\Sigma n(n - 1) = (40 \times 39) + (30 \times 29) + (20 \times 19) + (10 \times 9) = 1560 + 870 + 380 + 90 = 2900$ • $N(N - 1) = 100 \times 99 = 9900$ • $D = 9900 / 2900 = 3.41$ (accept 3.4)	3
05.2	• Woodland A has a higher biodiversity (greater variety and more even abundance) than woodland B • The index does not identify which species are present / a rare or important species may be missed; depends on identification skill and sampling method	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402; AQA spec 3.7.2.3.	
06.1	• t-test • The data are measured (continuous) values and two means are being compared • The data are (assumed to be) normally distributed	3
06.2	• Use the Mann-Whitney U test instead (compares medians)	1
06.3	• There is a 2% probability that the difference occurred by random chance • As this is less than 0.05 the difference is statistically significant / the null hypothesis is rejected	2
06.4	• The ranges of values (mean plus or minus SD) do not overlap: woodland up to 1.6 and clearing down to 2.4 • About 68% of values lie within one SD of each mean so the two data sets are clearly separated	2
06.5	• Significance shows the difference is unlikely to be due to chance, not that a causal relationship exists • Another variable (eg topography / aspect of the sites) could cause both; a controlled experiment or further evidence is needed	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 420-423; AQA spec 3.7.2.3.	
07.1	• Spearman's rank correlation coefficient • Mann-Whitney U test • Chi-squared test • t-test	4
07.2	• Chi-squared compares frequencies / counts in categories • Dissolved oxygen is continuously variable measured data, so a t-test or Mann-Whitney U test is needed	2
07.3	• There is always some remaining uncertainty / a probability that the result was produced by chance • Statistics only give a level of confidence (eg 95%, 99%) that the result is significant	2
07.4	• Bars drawn above and below each mean showing plus and minus one standard deviation • If the bars of two means overlap a lot the difference is probably not significant; if they do not overlap it probably is	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 422-423; AQA spec 3.7.2.3.	